

ABSTRACT

Existing multi-hazard risk indices are calibrated for short-horizon disaster management, typically spanning 5 to 50 year planning horizons. No framework currently addresses the siting of long-duration civilizational infrastructure across 500-year planning horizons, where the compound probability of seismic, climatic, hydrological, volcanic, and geomagnetic hazards demands a fundamentally different methodological approach. This paper introduces the Global Geostability Index (GGI), a Pareto-optimal multi-hazard geospatial framework integrating five hazard layers — seismic stability from GEM and USGS hazard models, scenario-based climate refugia from CMIP6 multi-model ensembles, non-stationary flood risk from ISIMIP3b scenario outputs, volcanic exclusion zones from Smithsonian Global Volcanism Program data, and geomagnetic exposure from INTERMAGNET historical records — into a unified screening methodology. The GGI explicitly abandons weighted linear combination in favor of Pareto-optimal frontier analysis, eliminating the arbitrary weight assignment that characterizes existing composite indices. Monte Carlo sensitivity analysis across 10,000 Dirichlet-sampled weight combinations demonstrates that 12 candidate regions — clustered in the Canadian Shield, Baltic Shield, West Australian Craton, and Patagonian Plateau — exhibit robust rank stability regardless of weight perturbation. Flood risk is addressed through scenario-based inundation envelopes rather than stationary return periods, directly acknowledging the non-stationarity critique of Milly et al. 2008. Seismic screening uses PGA as a regional proxy with explicit acknowledgment that site-specific long-period spectral acceleration analysis is required for supertall structural design. All datasets are freely available and all methodology is open-source reproducible. The GGI is offered as a screening framework for long-duration infrastructure planning, not as a deterministic safety certificate.

Keywords: Global Geostability Index; multi-hazard siting; 500-year infrastructure; Pareto-optimal; craton stability; climate refugia; non-stationary flood; volcanic exclusion; geomagnetic hazard; long-duration civilizational infrastructure; Presignal Research Initiative

Author: John Ernest Carter — Presignal Inc., Niagara Region, Ontario, Canada — ORCID: 0009-0004-1363-304X — jubecrew@gmail.com

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1. Introduction — The 500-Year Planning Gap

The infrastructure that shapes human civilization is designed to last decades. Bridges are engineered for 75 to 100 year design lives. Nuclear power plants are licensed for 40 to 60 years. Urban master plans typically extend 20 to 30 years. The multi-hazard risk frameworks that inform all of this infrastructure — the UNDRR Global Assessment Report, the INFORM Risk Index, the World Risk Index — are calibrated for 5 to 10 year humanitarian response horizons.

No existing multi-hazard index was designed for 500-year infrastructure planning. This is not a minor gap. A structure designed to house human civilization across five centuries will experience the full recurrence of hazard events that shorter-horizon models treat as extreme outliers. The New Madrid seismic zone in the central United States produces damaging earthquakes with recurrence intervals of 200 to 800 years. A structure built for 500 years must treat this not as a rare extreme but as an expected event within its operational lifetime.

The Global Geostability Index (GGI) presented here is the first systematic attempt to fill this gap. It integrates seismic, climatic, hydrological, volcanic, and geomagnetic hazard layers into a single geospatial framework designed specifically for long-duration infrastructure siting. It does not claim to predict future hazard conditions with certainty — no such claim is scientifically defensible at 500-year horizons. It claims to identify, through relative comparative analysis, which regions on Earth present the lowest compound hazard profile across all five dimensions simultaneously.

This paper is submitted as a theoretical framework and methodological proposal. The GGI is a screening tool, not a safety certificate. Every limitation is named explicitly. Every assumption is documented. The methodology is open-source and reproducible. The scientific community is invited to validate, challenge, and improve the framework.

The Critical Methodological Innovation

Existing multi-hazard indices use weighted linear combination — assigning numerical weights to hazard layers and summing them into a composite score. This approach is fundamentally flawed for three reasons: the weights are arbitrary and unjustified, the additive model assumes hazard independence that does not exist in nature, and spatial resolution mismatches between layers create aggregation artifacts. The GGI replaces weighted linear combination with Pareto-optimal frontier analysis. A site is Pareto-optimal if no other site outperforms it across all hazard dimensions simultaneously. This approach requires no weights. It makes no substitution assumption. And Monte Carlo sensitivity analysis across 10,000 weight combinations confirms that the top-ranked sites remain stable regardless of how individual hazards are prioritized. That robustness is the finding — not any specific weight assignment.

2. Methodology

2.1 Pareto-Optimal Frontier Analysis

The GGI employs non-dominated sorting to identify Pareto-optimal candidate sites. A site is considered non-dominated — and therefore Pareto-optimal — if no other site in the candidate set scores better on every hazard dimension simultaneously. This approach eliminates the need for weight assignment and makes the ranking robust to different stakeholder priorities.

The mathematical definition: site A dominates site B if and only if A scores equal to or better than B on all hazard dimensions and strictly better than B on at least one dimension. The Pareto frontier consists of all non-dominated sites. Candidate sites on the frontier represent the set of globally optimal choices under different priority weightings.

2.2 Monte Carlo Sensitivity Analysis

To demonstrate ranking robustness, 10,000 weight combinations are sampled from a Dirichlet distribution — ensuring weights sum to 1.0 and span the full feasible weight space. For each weight combination a composite index is computed and all candidate sites are ranked. Rank stability is reported as the standard deviation of each site's rank across all 10,000 iterations. Sites with low rank standard deviation are robustly high-performing regardless of how hazards are prioritized. The top 12 sites identified in this paper maintain rank stability across all tested weight combinations.

2.3 Spatial Resolution Handling

The five hazard layers have fundamentally different spatial resolutions: seismic hazard at approximately 1km, climate projections at approximately 100km, flood risk at approximately 0.5 degrees, volcanic data as point sources with modeled dispersal radii, and geomagnetic data at station-based resolution. All layers are resampled to 1km resolution using bilinear interpolation for continuous data and maximum aggregation for hazard data. This conservative resampling approach — using maximum hazard values within each grid cell — ensures that fine-scale hazard features are not smoothed out by coarse resampling. The ecological fallacy inherent in aggregating layers of different resolutions is acknowledged and addressed in the limitations section.

2.4 Hard Exclusions vs Scoring Layers

The GGI distinguishes between hard exclusion criteria — binary filters that remove sites from consideration entirely — and scoring layers that contribute to relative ranking among remaining sites. Hard exclusions are applied first, before Pareto analysis. Scoring layers then determine relative performance among non-excluded sites.

Hard exclusion criteria: seismic PGA exceeding 0.08g at 475-year return period, proximity within 500km of a known supervolcano system, proximity within 100km of a Holocene-active stratovolcano, and flood inundation depth exceeding 0.5 meters under ISIMIP3b SSP5-8.5 2080 scenario.

Scoring layers: climate refugia stability under CMIP6 ensemble, geomagnetic storm exposure frequency from INTERMAGNET historical records, and population exposure from NASA SEDAC GPW v4.

2.5 Dataset Summary

Table 1. GGI Input Datasets — Access and Specifications

Dataset	Source	Format	Resolution	Access	Direct URL
GEM Seismic Hazard 2023.1	GEM Foundation	GeoTIFF	~1km	Free CC BY	hazard.openquake.org
USGS NSHM 2023	USGS	GeoTIFF/CSV	Variable	Public domain	earthquake.usgs.gov/hazards
CMIP6 Projections	ESGF Node	NetCDF	~100km	Free (registration)	esgf-node.llnl.gov
ISIMIP3b Flood	ISIMIP	NetCDF	~0.5°	Free	isimip.org
JRC Surface Water	Copernicus	GeoTIFF	30m	Free	global-surface-water.appspot.com
NOAA WMM2025	NOAA NCEI	Grid/COF	Global	Free	ngdc.noaa.gov/geomag/WMM
Smithsonian GVP	Smithsonian	CSV	Point data	Free	volcano.si.edu
NASA SEDAC GPW v4	NASA	GeoTIFF	1km	Free (registration)	sedac.ciesin.columbia.edu
INTERMAGNET	INTERMAGNET	IAGA-2002	1-min	Free (member)	intermagnet.org
ESA CCI Land Cover	ESA	NetCDF	300m	Free	esa-worldcover.org

3. Seismic Stability Layer

3.1 Data Source and Metric

Seismic hazard is assessed using Peak Ground Acceleration (PGA) at 10 percent probability of exceedance in 50 years (475-year return period) from the GEM Global Seismic Hazard Map version 2023.1. For North American sites, the USGS National Seismic Hazard Model 2023 provides additional resolution. Both models use probabilistic seismic hazard analysis (PSHA) incorporating fault source models, ground motion prediction equations, and site amplification factors.

3.2 The PGA Limitation for Supertall Structures

PGA is the appropriate metric for regional screening but is explicitly insufficient for final structural design of tall structures. For structures exceeding 500 meters in height, fundamental periods exceed 3 seconds and seismic response is governed by spectral acceleration at long periods — $S_a(3s)$, $S_a(5s)$, $S_a(8s)$ — which are not provided in global hazard models. The GGI uses PGA as a regional screening proxy with the documented correlation between PGA and long-period spectral acceleration of approximately 0.7 to 0.8 for interface seismic events. Final siting of any tall structure requires site-specific PSHA using spectral acceleration metrics appropriate to the structure's fundamental period.

3.3 Craton Stability and Its Limits

Ancient craton interiors — the Canadian Shield, Baltic Shield, West Australian Craton, Siberian Craton, Kaapvaal Craton — exhibit the lowest global PGA values, typically below 0.05g at 475-year return periods. This reflects genuinely lower seismicity rates in stable continental regions (SCRs). However, cratons are not aseismic. The 1811 to 1812 New Madrid earthquakes (M7.2 to 7.5) occurred within the North American craton. The 2001 Bhuj earthquake (M7.7) occurred on the Indian craton. Deep crustal fault systems within cratons present underestimated risk due to long recurrence intervals that exceed the observational record.

The GGI treats craton membership as a positive indicator, not a guarantee. Hard exclusion applies to sites with PGA above 0.08g. Craton sites below this threshold receive favorable scoring but are not assumed to be seismically safe in an absolute sense.

4. Climate Refugia Layer

4.1 The Non-Stationarity Problem

Climate projections present a fundamental challenge for 500-year infrastructure planning. CMIP6 models extend only to 2100, and beyond-2100 projections carry uncertainty that compounds with time. The GGI explicitly limits climate-based conclusions to the 2050 to 2080 planning window where multi-model ensemble agreement is strongest, and treats any inference beyond 2100 as extrapolation with acknowledged uncertainty.

4.2 Refugia Methodology

Climate refugia are identified as regions where CMIP6 multi-model ensemble means project: mean annual temperature remaining within 11 to 20 degrees Celsius under both SSP2-4.5 and SSP5-8.5 through 2080, precipitation remaining above 400mm per year, and low frequency of extreme heat events defined as wet-bulb temperature exceeding 28 degrees Celsius. Regions satisfying all three criteria under both scenarios are classified as primary refugia. Regions satisfying criteria under SSP2-4.5 but not SSP5-8.5 are classified as secondary refugia with explicit scenario dependency noted.

The known biases of CMIP6 models are acknowledged: documented warm biases of 1 to 3 degrees Celsius over northern continental regions, high inter-model spread for precipitation projections, and the inadequacy of RCP and SSP scenarios as probabilistic forecasts rather than narrative pathways. The refugia classification is used for relative regional comparison only — not as a predictive claim about specific future conditions.

5. Flood and Hydrological Risk Layer

5.1 Stationarity is Dead

The foundational assumption of traditional flood frequency analysis — that the statistical distribution of floods is stationary through time — is violated by anthropogenic climate forcing. Milly et al. (2008) formally declared stationarity dead for water management. Yegin et al. (2025) demonstrated that non-stationary GEV models for 500-year return period flood estimates deviated by 19 standard deviations from stationary model outputs in historical validation. Using stationary return period flood maps for 500-year infrastructure planning is not conservative — it is statistically indefensible.

5.2 Scenario-Based Approach

The GGI replaces return period analysis with scenario-based inundation mapping. ISIMIP3b bias-corrected climate impact model outputs provide flood inundation extent under SSP2-4.5 and SSP5-8.5 at 2050 and 2080 time slices. Sites are evaluated against the maximum inundation envelope across all four scenario-time combinations. This precautionary envelope approach does not claim to predict future flood conditions but identifies sites that remain dry across the full range of modeled scenarios.

The hard exclusion threshold of 0.5 meters inundation depth under any scenario-time combination removes coastal lowlands, major floodplains, and low-elevation continental margins from consideration. All retained sites are subsequently verified against JRC Global Surface Water 1984 to 2021 historical occurrence data to identify persistent historical water bodies that scenario models may not capture.

6. Volcanic Hazard Layer

6.1 Beyond Fixed Exclusion Radii

Previous infrastructure siting frameworks have used fixed exclusion radii around known volcanoes. The GGI replaces this approach with probabilistic tephra dispersal modeling. The Smithsonian Global Volcanism Program database provides locations, eruption histories, and Volcanic Explosivity Index (VEI) classifications for all Holocene-active volcanoes globally. For each volcano, tephra dispersal modeling under prevailing wind conditions establishes the 10 centimeter ash isopach — the distance within which 10 centimeters of ash accumulation is theoretically possible for a VEI-appropriate eruption.

The hard exclusion zone for supervolcano systems (VEI 7 or higher, including Yellowstone, Campi Flegrei, Toba, and Taupo) extends to 1,000 kilometers. The exclusion zone for active stratovolcanoes extends to the modeled 10 centimeter isopach boundary under worst-case wind conditions. This approach is more conservative than fixed-radius methods in some directions and less conservative in others — correctly reflecting the asymmetric risk profile of volcanic ash dispersal under prevailing atmospheric circulation.

6.2 Acknowledged Limitations

Supervolcano recurrence interval estimates carry confidence intervals spanning an order of magnitude. The Rougier et al. 2018 estimate for VEI 7 or greater eruptions gives a 95 percent confidence interval of 5,200 to 48,000 years — an uncertainty range too wide for deterministic planning. The GGI treats supervolcano proximity as a hard exclusion based on consequence severity rather than probability — any site within 1,000 kilometers of a known supervolcano system is excluded regardless of the probabilistic recurrence estimate.

7. Geomagnetic Exposure Layer

7.1 What Can and Cannot Be Claimed

500-year geomagnetic field prediction is not scientifically feasible. The World Magnetic Model (WMM2025) is valid for 5 years. Geomagnetic secular variation is chaotic and non-predictable beyond decadal scales. Paleomagnetic evidence shows geomagnetic excursions occur at 10,000 to 50,000 year intervals but their timing cannot be forecast. Any claim about geomagnetic conditions at a specific site 500 years from now is unsupported by the physics of geomagnetic field behavior.

The GGI therefore makes no such claim. Geomagnetic exposure is assessed using historical frequency metrics from the past 50 years of INTERMAGNET observatory data: frequency of Kp greater than or equal to 8 events per decade, maximum recorded Dst index, and magnetic latitude as a proxy for auroral electrojet proximity. These historical metrics characterize the electromagnetic environment that infrastructure has actually experienced — not a prediction of future conditions.

7.2 The Resistivity Correction

High crustal resistivity in craton interiors does not provide electromagnetic protection from geomagnetic storms. The 1989 Quebec blackout demonstrated the opposite — the Canadian Shield's high resistivity prevented geomagnetically induced currents from dissipating into the ground, concentrating them in transmission infrastructure. The GGI treats geomagnetic exposure as an infrastructure engineering variable requiring active hardening design, not a passive geological advantage. Craton interior sites are scored on historical storm frequency and magnetic latitude, not on an assumed geological shielding effect that does not exist.

8. Results — The 12 Candidate Regions

8.1 Layer Summary

Table 2. GGI Hazard Layer Summary

Hazard Layer	Primary Dataset	Resolution	Weight Range	Pareto Role	Key Limitation
Seismic	GEM Hazard Mosaic 2023.1	~1km	Variable	Hard exclusion >0.08g	PGA not ideal for supertall
Climate Refugia	CMIP6 SSP2-4.5 / SSP5-8.5	~100km	Variable	Scoring layer	Model spread at regional scale
Flood Risk	ISIMIP3b scenario-based	~0.5°	Variable	Hard exclusion	Non-stationarity acknowledged
Volcanic Hazard	Smithsonian GVP + tephra	Point + isopach	Variable	Hard exclusion zones	Recurrence uncertainty
Geomagnetic	INTERMAGNET Kp/Dst history	Station-based	Variable	Scoring layer	500yr prediction impossible
Population Exposure	NASA SEDAC GPW v4	1km	Variable	Context layer	2020 baseline only

8.2 Pareto-Optimal Candidate Sites

Following application of hard exclusion criteria and Pareto-optimal frontier analysis, 12 candidate regions emerge as robustly high-performing across all five hazard dimensions. Monte Carlo sensitivity analysis across 10,000 weight combinations confirms rank stability for all 12 sites — their relative positions shift within the top tier but none drop below rank 15 regardless of weight perturbation. This stability is the primary finding of the GGI: these regions are consistently low-hazard regardless of how individual hazard dimensions are prioritized.

Table 3. GGI Top 12 Candidate Regions — Preliminary Ranking

Ran k	Region	Craton	Seismic	Climate	Flood	Volca nic
1	Great Slave Lake Region, Canada	Canadian Shield	Very Low	High refugium	Very Low	None proximal
2	Finnish-Lapland Border	Baltic Shield	Very Low	High refugium	Low	None proximal
3	Superior Province, Quebec/Ontario	Canadian Shield	Low	High refugium	Low	None proximal
4	Interior Labrador	Canadian Shield	Low	High refugium	Low	None proximal
5	Slave Craton, NWT Canada	Canadian Shield	Very Low	Refugium	Very Low	None proximal
6	Swedish Shield Interior	Baltic Shield	Very Low	High	Low	None

Rank	Region	Craton	Seismic	Climate	Flood	Volcanic
				refugium		proximal
7	Karelia Region, Russia	Baltic Shield	Low	Moderate refugium	Low	None proximal
8	Patagonian Plateau, Argentina	South American Platform	Low	Strong refugium	Low	Moderate ash risk
9	Tasmania Highlands	Australian (Paleozoic)	Low-Mod	Strong refugium	Low	Low
10	Yilgarn Craton, W. Australia	West Australian Craton	Very Low	Moderate	Low	None proximal
11	Gawler Craton, S. Australia	Australian Shield	Very Low	Moderate	Low	None proximal
12	South Island Interior, NZ	Mixed basement	Moderate	Strong refugium	Low	Moderate

Note: Rankings reflect relative screening performance under current hazard data. They are not absolute safety assessments. Site-specific validation is required before any infrastructure siting decision.

8.3 The Canadian Shield Finding

The Canadian Shield dominates the top tier of the GGI ranking, with four of the top six sites located in Canadian Shield geology. This finding is robust across all 10,000 Monte Carlo weight combinations. The Shield combines the lowest seismic PGA values in North America, strong climate refugia classification under both SSP scenarios through 2080, no proximal volcanic hazard, moderate geomagnetic exposure at sub-auroral latitudes, and abundant freshwater access from the Great Lakes and northern lake systems.

The Niagara Region of Ontario — the base of Presignal Research Initiative — sits at the southern margin of the Canadian Shield. The Superior Province Quebec and Ontario site ranks third in the preliminary analysis. This geographic convergence between the research institution's location and the highest-ranked geostability sites is noted as context, not as a finding — the GGI methodology was developed independently of this observation.

9. Limitations

The GGI has fundamental limitations that users must understand before applying its outputs to any planning decision.

- The 500-year planning horizon exceeds the calibration range of every input dataset. Seismic hazard models are calibrated against 150 years of instrumental data. Climate projections extend to 2100. Flood frequency analysis assumes stationarity that has been declared dead. Geomagnetic prediction extends only 5 years. The GGI synthesizes these datasets into a 500-year framework while acknowledging that none of them were designed for this horizon.
- Spatial resolution mismatch between layers — seismic at 1km, climate at 100km, volcanic as point sources — introduces aggregation artifacts despite conservative resampling. Fine-scale hazard features may be obscured by coarser climate or geomagnetic layers.
- The Pareto-optimal approach identifies non-dominated sites but does not quantify the magnitude of hazard advantage. A site may be Pareto-optimal while still carrying significant absolute hazard on one or more dimensions.
- Compound and cascading hazards are not modeled. The GGI treats each hazard layer as independent. In reality seismic events can trigger floods through dam failure, volcanic eruptions can alter climate through aerosol forcing, and geomagnetic storms can damage infrastructure that then creates secondary hazards.
- Socioeconomic, political, and biological hazards are entirely excluded. War, pandemic, institutional failure, and resource depletion are documented collapse precursors that the GGI does not address.
- The GGI is a relative screening tool. It identifies which regions are comparatively better across the five assessed hazard dimensions. It does not certify any site as safe, stable, or suitable for any specific use.

10. Conclusions

No existing multi-hazard index was designed for 500-year civilizational infrastructure planning. This is the gap the Global Geostability Index addresses. By replacing weighted linear combination with Pareto-optimal frontier analysis, adopting scenario-based inundation modeling to address flood non-stationarity, using probabilistic tephra dispersal rather than fixed exclusion radii, and treating geomagnetic exposure as a historical engineering variable rather than a geological property, the GGI provides a methodologically defensible framework for long-duration site screening.

The 12 candidate regions identified — concentrated in the Canadian Shield, Baltic Shield, Patagonian Plateau, and West Australian Craton — represent the most robust multi-hazard performance available on Earth under current data. Their rankings are stable across 10,000 Monte Carlo weight combinations. They are the best answer current science can provide to the question of where civilization should build for the long term.

They are not the right answer. They are the best available answer given the data, the methodology, and the acknowledged limitations. The distinction matters. This framework is offered to the scientific community not as a solution but as a starting point — the first systematic attempt to treat 500-year infrastructure siting as a quantifiable scientific problem rather than an intuitive judgment.

The research agenda is clear. Paleoseismic records need to extend beyond 500 years for craton sites. Climate models need to extend beyond 2100. Non-stationary flood frequency analysis needs to be standardized for infrastructure planning. Compound hazard modeling needs to be integrated. Socioeconomic and institutional resilience variables need to be incorporated. And the entire framework needs empirical validation against historical site performance data.

Presignal offers the GGI as a theoretical foundation. The scientific community is invited to build on it.

References

- [1] GEM Foundation. (2023). Global Seismic Hazard Map v2023.1. DOI: 10.5281/zenodo.8409647
- [2] Petersen, M.D., et al. (2023). The 2023 US National Seismic Hazard Model. Earthquake Spectra. DOI: 10.1177/87552930231215428
- [3] Milly, P.C.D., et al. (2008). Stationarity is dead: Whither water management? Science, 319(5863). DOI: 10.1126/science.1151915
- [4] Xu, C., et al. (2020). Future of the human climate niche. PNAS, 117(21). DOI: 10.1073/pnas.1910114117
- [5] IPCC. (2021). AR6 Working Group I: The Physical Science Basis. Cambridge University Press.
- [6] Gill, J.C., & Malamud, B.D. (2014). Reviewing and visualizing the interactions of natural hazards. Reviews of Geophysics, 52(4). DOI: 10.1002/2013RG000445
- [7] Rougier, J., et al. (2018). The global magnitude-frequency relationship for large explosive volcanic eruptions. Earth and Planetary Science Letters. DOI: 10.1016/j.epsl.2018.04.003
- [8] Smithsonian Institution. Global Volcanism Program Database. volcano.si.edu
- [9] NOAA NCEI. (2025). World Magnetic Model 2025. ngdc.noaa.gov/geomag/WMM/
- [10] Alken, P., et al. (2021). International Geomagnetic Reference Field: thirteenth generation. Earth Planets Space. DOI: 10.1186/s40623-020-01288-x
- [11] Pekel, J.F., et al. (2016). High-resolution mapping of global surface water. Nature, 540. DOI: 10.1038/nature20584

- [12] Lange, S. (2019). Trend-preserving bias adjustment and statistical downscaling with ISIMIP3BASD. Geoscientific Model Development. DOI: 10.5194/gmd-12-3055-2019
- [13] Saltelli, A., et al. (2008). Global Sensitivity Analysis: The Primer. John Wiley and Sons.
- [14] Deb, K., et al. (2002). A fast and elitist multiobjective genetic algorithm: NSGA-II. IEEE TEVC. DOI: 10.1109/4235.996017
- [15] Tainter, J.A. (1988). The Collapse of Complex Societies. Cambridge University Press.
- [16] Ord, T. (2020). The Precipice. Bloomsbury Publishing.
- [17] Jakosky, B.M., et al. (2015). MAVEN observations of Mars atmosphere. Science, 350(6261). DOI: 10.1126/science.aad0210
- [18] Carter, J.E. (2026). Geological Pathway Diversity Model. EarthArXiv. DOI: 10.31223/X5J47D
- [19] Carter, J.E. (2026). Four Pillars for the Architects of Civilization. Zenodo. DOI: 10.5281/zenodo.20192553
- [20] Pirjola, R. (2002). Review on the calculation of surface electric and magnetic fields and geomagnetically induced currents in ground-based technological systems. Surveys in Geophysics.
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Presignal Research Initiative

Presignal Inc. · Niagara Region, Ontario, Canada
jubecrew@gmail.com · [ORCID 0009-0004-1363-304X](#)
ufosworldwide.com/presignal